OpenSourceRail v0.5

Ferrite PM maglev — primary design reader edition

Fixed magnetic modules · compact elevated guidebeams · Rust controls · local manufacture

Provisional digital studies. Complete vehicle, civil and software qualification remains open. Accepted physical releases: 0.

OpenSourceRail v0.5 — ferrite PM maglev

docs/open-source-rail-overview.md

Fewer moving parts, local manufacture and compact elevated urban routes.



OpenSourceRail maglev station concept

The primary design uses fixed onboard ferrite magnetic modules and powered linear electromagnetic propulsion, replacing conventional wheelsets, axles, gearboxes and rotating traction motors in normal running gear. Windings, drives, cooling, sensors, protection, passenger doors and emergency systems remain necessary. Contactless running aims to reduce wheel/rail contact noise and wear; acoustic performance and whole-system maintenance still require measurement.

Simple civil works and local capability

The research target is one compact integrated concrete guidebeam per direction, with necessary ferromagnetic reaction surfaces, repeatable supports and no continuous external walkway. A vertical spiral evacuation chute is the preferred alternative under investigation. Walkway removal has not been approved. Route clearance, safe landings, assisted evacuation, stopping/recovery and fire response must work at every permitted stopping location.

The 99% elevated-mainline objective and straight tangents aim to reach constrained urban corridors with less ground occupation and disruption. Land rights, utilities, access, wildlife, waterways, curvature and foundation feasibility still govern each route. No universal clearance or zero-disruption claim is made.

Local casting, steel fabrication, cartridge assembly, winding, vehicle structures, interiors, wiring, Rust software and integration can create domestic capability. The complete local-content percentage, installed programme cost and funding requirement remain unknown until origin-traced bills, processes and supplier offers exist. Costs use the current material quantities, manufacturing routes and explicit planning rates. Unpriced scope remains open; no supplier quotation is implied.

Inspect and reproduce the current work

Current digital capability	Scope
City planning	266 separate primary maglev packages; retained geography/demand, recalculated formation, duty, fleet and charging
Magnetic interface	624 native solids; partial lift/cartridge/frame geometry
Beam-only civil kit	54 native solids; selected geometry without external walkways
Evacuation entry and support	20 partial entry parts and 3 support envelopes; not complete manufactured products
Controls	Allocation-free Rust core, 24-module replay, I/O contracts and advisory evacuation state machine
Simulation and media	Field equation controls, actual two-way Rust/beam replay, five assembly visibility videos, native images and source receipts
Engineering acceptance	0 accepted physical releases; design, manufacture and operational gates remain open

Use `./osr build` to verify dependencies and rebuild stale implemented studies. `./osr workbench` opens the primary maglev UI. `./osr design check` independently regenerates the report. `./osr book` builds the current reader edition. The scheduled runner repeats dependency verification after source changes; it does not invent missing physical tests or certify engineering completion.

Open the current Workbench, review calculations, CAD and videos, read the implementation and open work, or download the primary reader book.

OpenSourceRail v0.5 ferrite-maglev engineering

engineering/ferrite_maglev/README.md

The current design uses ferrite PM-biased active-attraction support, fixed onboard windings and electronics, passive reaction iron and compact concrete guidebeams. Rust control packages run offline models for eventual off-the-shelf SBC implementation. Powered linear propulsion, cooling, sensing, independent protection, braking, capture, parking and recovery remain necessary. Accepted engineering and physical releases: zero.

Run and inspect

```
./osr build
./osr design run
./osr design check
./osr design verify
./osr city baghdad
./osr sim baghdad --cycles 2
./osr workbench
./osr test
./osr book
cargo test --locked -p osr-maglev-control --all-targets
```

Generated report · Workbench · city directory · architecture.

`./osr capabilities` publishes executable sources and their scope. `./osr catalogue plan --city baghdad` resolves a selected regeneration, and `./osr catalogue run --city baghdad` rebuilds that city. `./osr readiness` reports current-source gates. `./osr design release-check` remains held while acceptance is open.

The dependency runner verifies implemented native, numerical, control, city and publication prerequisites in order. It rebuilds stale evidence and holds dependent outputs after failures. An interprocess lock excludes concurrent timer/manual builds. Runtime progress is stored under `build/maglev-workflow/latest.json`. This is a finite reproducible build of implemented evaluators; missing physical tests and engineering decisions require additional work.

Current inputs and design records

Concern	Source
Architecture	Infrastructure policy
Vehicle and civil configuration	Reference configuration, configuration reader, study assumptions
Requirements and work allocation	Requirements data, current systems register
Cost quantities and manufacturing	Process model, planning rates
Magnetic/native geometry	Geometry, field equations, native export
City service and geography	Each city's `city.json`; compiler
Control	Rust maglev package, offline SBC profile

Baghdad uses six 17 m bodies with shared coupling, platform, magnetic-arm and boarding-gap assumptions. The partial station model includes floor frames, guidebeams, island platform and a centred flush lift floor. It omits complete vehicle and station equipment; the boarding gap requires a qualified bridge plate. Native exports identify the exact modeled revision.

Implementation and closure status

Capability	Implemented scope	Outstanding evidence
Systems register	Current system items, requirement relationships, work packages, manufacturing and test obligations	Approved quantitative criteria and accepted exact-configuration evidence
Native geometry	Partial magnetic interface, civil bay, beam-only variant, evacuation entry, support envelopes and Baghdad station; FreeCAD/STEP exports, reopening checks, images and short visibility videos	Full production geometry, qualified materials, joints, loads, tolerance chains and manufacturing drawings
Magnetic field	2D variable-reluctivity equation, manufactured-solution controls and explicit nonlinear material hypotheses	Coupon force convergence, measured material data, 3D force/loss maps and physical rig correlation
Rust control	Module allocation, five-mode virtual replay, protection interfaces, support-preserving stops and advisory evacuation	Actual actuator/sensor model, isolated physical I/O, independent protection and measured timing
Coupled replay	Two-way exchange of Rust module forces and flexible vertical-beam deflections	Flexible vehicle, lateral/torsional/joint/soil dynamics, moving loads and correlated stability/ride/fault measurements
City planning	Capacity-matched formations, duty, fleet, charging, guideway quantities, provisional costs and manufacturing process needs	Located terminal loops, route/ground/utility surveys, recovery, full duty and complete programme economics
Industrialisation	Current material quantities and explicit process/rate assumptions	Configured suppliers, qualified processes, origin-traced MBOM, production capacity, logistics and installed scope

Assembly videos show final-position visibility sequences. They do not simulate motion, collision-free erection, crane operation or chute deployment. CAD export integrity does not establish strength or manufacturing readiness. Virtual replay steps do not demonstrate physical deadlines. Equation residuals do not establish real lift capacity.

Manufacturing and costs

Cost estimates start from current maglev quantities and fabrication operations. The reference formation is six cars. Per-car equipment, shared train equipment, beam/support quantities, station and factory scope must be kept distinct. Planning rates are explicit assumptions, rather than supplier quotations. A missing first-pilot/NRE total stays unpriced. Project scope, contingency, lifecycle and finance require their own evidence.

City `engineering/maglev/manufacturing.json` records the materials, processing, equipment needs and construction approaches behind its estimate. Candidate processes include controlled precast casting/curing and inspection, reinforcement/prestress work, reaction-iron fabrication and fitting, structural cutting/joining, coil winding, cartridge assembly, wiring, software integration and factory testing. Transport, temporary works, crane radius/capacity, joints and site acceptance remain part of installed delivery.

Prefer verified British manufacture when considering UK suppliers. A UK distributor or integrator does not establish UK manufacturing origin. Likewise domestic assembly does not change imported cell, semiconductor, sensor or raw-material origin. Complete local-content percentage remains unknown until the complete bill and origin evidence exist.

Evacuation and compact civil works

The research target is one integrated beam per direction without a continuous external walkway. Necessary reaction iron, supports, foundations and connections remain. A vertical spiral controlled-descent chute is the preferred unqualified onboard candidate. The evacuation study tracks variable height, anchorage, ground landing, assistance, fire/weather, partial failure, rescue, lifecycle and driverless operation. Walkway removal is conditional on accepted route and onboard alternatives at every permitted stopping location.

Implementation roadmap

The closure queue covers complete mechanical/civil geometry, magnetic qualification, dynamics, physical controls, origin-traced industrialisation, city site evidence and demonstrator acceptance. Requirements allocation and successful digital checks define progress within their scope; neither constitutes a manufacturing or operating release.

osr-maglev-control

crates/osr-maglev-control/README.md

Portable v0.5 maglev study core, using Rust `no_std`, no heap and no external dependencies. It targets simulation and off-the-shelf SBC replay. Every computed output has `physical_output_authorised = false`; the I/O boundary records study outputs and supplies no current, gate-drive or PWM operation.

The existing `evaluate(previous, inputs, now_ns, params)` API remains a single-module integer current hypothesis. Its polarity, damping, envelope, time/replay, stop/reset and support-power tests remain compatible. Increasing an upward-attraction gap asks for increasing support. The controller cannot be reused unchanged for shaped-pole passive vertical support.

24-module force and moment study

`array::evaluate_array` adds coupled heave, roll, pitch, lateral displacement and yaw feedback and `array::allocate` distributes requested forces and moments over 24 stations. This is one rigid car, not a three-car coupled train or a validated six-degree-of-freedom suspension. Longitudinal force has no actuator in this model; a nonzero longitudinal demand appears explicitly as unachieved residual. Propulsion, braking, roll/yaw cross-coupling from real traction, flexible car/beam modes, nonlinear finite-angle dynamics and contact dynamics remain open.

Coordinates use x forward, y left and z up, with the assumed car centre of mass as origin. Station positions and gaps are metres, force is newtons, moments are newton-metres, small angles are radians and timestamps are monotonic nanoseconds. Upward force is positive attraction. Lateral force is positive left. Each moment is the sum of `position cross force`. The explicit seed geometry is 12 paired stations at x = -5.5, -4.5, …, 5.5 m and y = ±0.88 m, z = 0 m; this matches the candidate 12 m magnetic strips. Actual centre of mass, attachment heights, lever arms, tolerances and dynamic mass properties need physical closure.

Each station has a bounded upward channel and a signed lateral channel. The illustrative limits are 0–20 kN upward, ±4 kN lateral and 2 MN/s slew per channel. These numbers are hypothetical force envelopes, not measured magnetic maps, current/voltage/thermal limits or proved simultaneous lift/guidance capability. A real PM-biased module can retain force after drive loss: a failed station takes an explicit specified residual force. Zero in the example is an ideal failure assumption. Unknown failed-channel forces require a separate bounded uncertainty and protection model before hardware use.

The allocator uses at most 256 bounded coordinate sweeps to minimise squared force residuals and moment residuals scaled by a declared 6 m lever arm. It reports the independently reconstructed resultant, residual wrench, constraint limits and feasibility tolerances. A solver iteration count establishes no physical deadline. Infeasible requests remain visible; no six-axis success is inferred from solver completion. Slew limits apply against a prior force frame; without one the first study allocation has no force history and applies only static limits. This is not a physical startup or touchdown procedure.

The mode estimator fits a small-angle rigid plane to 24 dual-gap samples and fits lateral displacement/yaw to 24 dual lateral samples. Every individual gap and lateral sample is checked before averaging; disagreement, nonfinite values and degenerate geometry fail. This fit is not a flexible-body observer. Sensor channels share one supplied acquisition timestamp; per-channel skew, physical independence, common-cause failure and calibration are not established. Derivatives use acquisition time. Host time/sequence checks reject stale, future, replayed, regressing and late frames.

A stop request preserves fresh bounded support. A station failure, active bound or infeasible allocation latches a traction stop while available ideal support remains active. Missing valid sensing or power requests external local fallback, never frozen study forces. A prior fault inhibits traction until a fresh healthy frame, reported stopped/secured conditions and reset coincide. Capture, fallback, independent protection energy and secure-condition verification need separate engineering; they are not supplied by these Boolean reports.

Reproducible offline checks

Run:

```
cargo test -p osr-maglev-control
cargo clippy -p osr-maglev-control --all-targets -- -D warnings
cargo run -p osr-maglev-control --example sbc_replay
cargo run -p osr-maglev-control --example multimodule_replay
```

The array replay integrates ten seconds of virtual 1 ms frames, prints CSV at 500 ms intervals and asserts feasibility and a final small five-mode error. Its 27,750 kg AW2 mass is the profile's 18,750 kg matched tare plus 120 × 75 kg passengers. Gains, diagonal inertias, force limits and instantaneous actuator response are synthetic. It starts with coupled displacement/tilt, commands a stop at 0.5 s and loses station 0 at 1 s under a specified zero-residual failure hypothesis. It keeps fresh support and the stop latch active. This trajectory is a numerical example, not a closed-loop stability or ride-quality

proof.

`ideal::IdealPlant` integrates a constant wrench exactly within each small-angle step. Tests independently check uniform equilibrium (272,227.5 N weight), moment signs, force/moment reconstruction, five-component demand, failed-force redistribution, capacity and slew infeasibility, all-failed residuals, sensor and time rejection, support-preserving stops and secure reset. Mechanical interface power agrees with resultant wrench power, and one-step work agrees with change in kinetic plus gravitational energy. These conservation checks validate the ideal mechanics arithmetic, not actual magnet energy or electrical losses.

`io::StudyIo`, `host_tick` and `ReplayIo` define an acquisition/record boundary without owning an OS clock or hardware. Recording/acquisition failures return to the host; the host must engineer its local fallback. Repeated single-frame stub reads are rejected as replay. No scheduler, GPIO, ADC or network transport is validated, and this example measures neither SBC timing nor I/O latency.

Physical gates remain open: calibrated nonlinear magnetic force/loss maps over gap/current/temperature and shared flux, identified electrical/thermal and mechanical dynamics, actuator failure residuals, independent sensors/protection, measured end-to-end target timing, multi-module rigs, flexible guideway and loaded vehicle tests, braking/capture and train/route acceptance. No operational or manufacturing authority is inherited from the numerical study.

Flexible guideway numerical demonstrator

`coupled_bridge` connects this exact array core to the separately labelled Python beam demonstrator in `engineering/ferrite_maglev/coupled.py`. From the repository root, run:

```
cargo build -p osr-maglev-control --release --example coupled_bridge
.venv/bin/python -m engineering.ferrite_maglev.coupled --duration-s 3
.venv/bin/python -m pytest -q engineering/analysis/tests/test_ferrite_maglev_coupled.py
```

The Python host sends all 24 local relative gaps each virtual millisecond. Rust returns bounded support forces, which load a 25 m simply supported Euler–Bernoulli beam; its local displacement feeds the next sensor frame. This is two-way offline coupling, with no physical output. The beam assumes E = 30 GPa, I = 0.193624 m⁴, line mass 1750 kg/m and 2% modal damping, retaining six vertical modes of a 50-element consistent-mass model. Independent point-load closed forms check static response; analytical pinned-beam frequencies and virtual-work/undamped-energy checks control the numerical implementation.

The model can show global beam displacement exceeding 8 mm while the car follows that displacement and the relative gaps remain inside the example's study envelope. This establishes neither a gap requirement nor structural acceptance. It omits dead-load camber, torsion/corbel flexibility, moving/train loads, flexible car structure, soil, joints, fatigue, real magnetic/electrical dynamics and independent protection. The line protocol accepts at most 20,000 frames, 8 KiB per frame, and measures no hardware timing. The three-mode car motion and vertical beam coupling do not provide a six-degree-of-freedom stability proof.

Offline evacuation sequencing

`egress::evaluate` adds allocation-free doorway sequencing with an exact declared candidate/door context, fresh monotonic frames, explicit readiness/assistance prerequisites, latched contradiction/context/deadline faults, and secured/stowed reset. It reports study traction inhibition and continued-support obligations; it supplies no door, restraint, chute or support output adapter. Input assertions are unauthenticated study facts, not verified independent sensor or safety evidence.

`cargo run -p osr-maglev-control --release --example egress_replay -- REVISION PATH_CLEAR HEIGHT_COVERED` prints a bounded nine-frame synthetic sequence. The linked Python study supplies its candidate tag and calculated hypothetical path/height gates, and retains negative controls. Loss of prerequisites after deployment begins holds the sequence; no automatic retraction is commanded. The egress/support composition test preserves fresh support control while inhibiting traction. Physical timing, independent protection, accessible passenger evacuation and railway approval remain open.

OpenSourceRail architecture

docs/ARCHITECTURE.md

OpenSourceRail v0.5 develops ferrite PM-biased active-attraction maglev for urban transport. Its current physical basis is fixed onboard magnetic modules, powered linear propulsion, passive reaction iron and compact elevated concrete guidebeams. Batteries and station charging support operation; independent protection, cooling, capture, braking, parking and recovery remain essential design obligations.

The objective is broad city access with a small ground footprint, few moving running-gear parts and practical local manufacture. Ordinary mainlines target repeated elevated bays. Route rights, foundations, utilities, curvature, station access and evacuation determine where the concept is feasible.

Linked design layers

Layer	Controlled definition	Evidence
Service and city	Geography, demand, capacity, headways, route lengths, station locations and local assumptions	Formation, fleet, energy/charging, quantities, provisional costs and unresolved route gates
Mechanical	Magnetic cartridges, structural frame, passenger systems, energy, doors and evacuation	Partial native solids, mass budgets, interface checks, analytical screens and production obligations
Civil	Guidebeams, reaction iron, joints, supports, foundations, stations and services	Section/beam calculations, support-demand studies, native envelopes and manufacturing/erection quantities
Software	Rust support control, protection interfaces, module allocation, fault handling and advisory evacuation	Offline executable tests, virtual replay and a two-way flexible-beam coupling
Industrialisation	Materials, fabrication processes, tooling, factory throughput, transport, erection and origin	Quantity-and-process planning allowances and explicit unpriced scope
Acceptance	Exact revision, requirements, tests, evidence, reviewer and serial identity	Source-bound receipts and open engineering/manufacturing/physical gates

Infrastructure policy, configuration and engineering modules control these interfaces. Generated city inputs use `city.json`; their outputs follow the same maglev compiler and cost model.

Current evidence boundary

Native CAD covers selected partial assemblies. Assembly visibility videos reveal final-position geometry; they do not establish collision-free installation or deployment. Field-equation controls do not establish real magnetic lift capacity. Virtual control/beam replay does not establish physical SBC deadlines or qualified train dynamics. Open geographic data supports planning estimates, rather than surveyed boundaries, soil capacity or utility clearance.

A continuous-walkway-free guideway remains conditional on qualified onboard and route evacuation. Vertical spiral chutes require usable landings and assisted evacuation at every allowed stopping location. No blanket waiver of evacuation or structural requirements is implied by the compact layout.

All release flags remain false and accepted engineering/physical releases remain zero. Local content, installed programme cost and the funding requirement require complete origin-traced quantities, configured supplier offers, construction scope and financing terms. Successful publication integrity is separate from engineering acceptance.

Roadmap

docs/ROADMAP.md

OpenSourceRail v0.5 uses one ferrite-maglev architecture across mechanical, civil, software, manufacturing and city planning. Implemented work and commands and the current report identify the tested digital scope. Accepted engineering and physical releases remain zero.

Primary maglev closure queue

Package	Evidence required for closure
Requirements and configuration	Approved loads, service envelope, tolerances and quantitative acceptance criteria for exact revisions
Magnetic assemblies	Complete support/guidance/propulsion/capture assemblies, connections and dimensional chains
Magnetic force and loss	Converged force calculations, qualified material curves, 3D winding/end effects, thermal maps and correlated rig measurements
Vehicle	Complete body, doors, interiors, storage, cooling, controls and evacuation; mass, centre of gravity, inertia and structural qualification
Civil kit	Beams, reaction iron, connections, supports, foundations, stations, services and all handling/construction stages
Control and protection	Qualified plant model, independent protection, loss-of-power behavior, braking, capture and recovery interfaces
Coupled dynamics	Moving trains, flexible bodies, lateral/torsional modes, joints and soil with calibrated models
SBC hardware	Isolated I/O, power stages, measured deadlines, watchdogs and independent protection on selected equipment
City delivery	Located routes, terminal loops, platforms, charging, storage/recovery, all-stop evacuation and site investigations
Manufacturing and economics	Complete origin-traced MBOM, process routes, machine/tooling options, throughput, transport, erection, maintenance and financing
Demonstrator	Traceable manufactured units, independent inspections and physical acceptance on identified serials and revisions

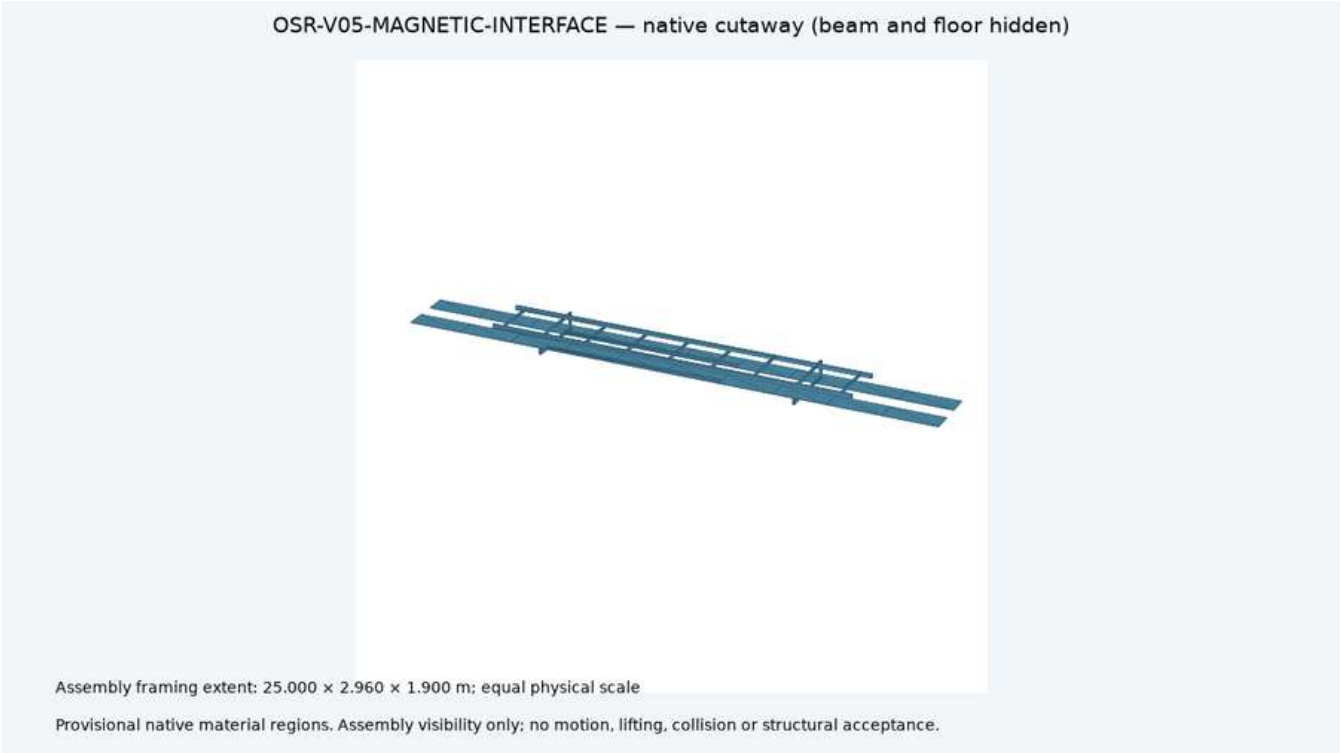
The automated runner can regenerate the implemented studies and hold downstream evidence after failures. Missing evaluators, measurements and approvals remain open work, even after every current automated check passes.

Native geometry review



Partial six-car Baghdad station, magnetic guidebeams and island lift; acceptance remains open.

Partial six-car Baghdad station, magnetic guidebeams and island lift; acceptance remains open.



Partial fixed magnetic interface; not a complete vehicle.

Partial fixed magnetic interface; not a complete vehicle.



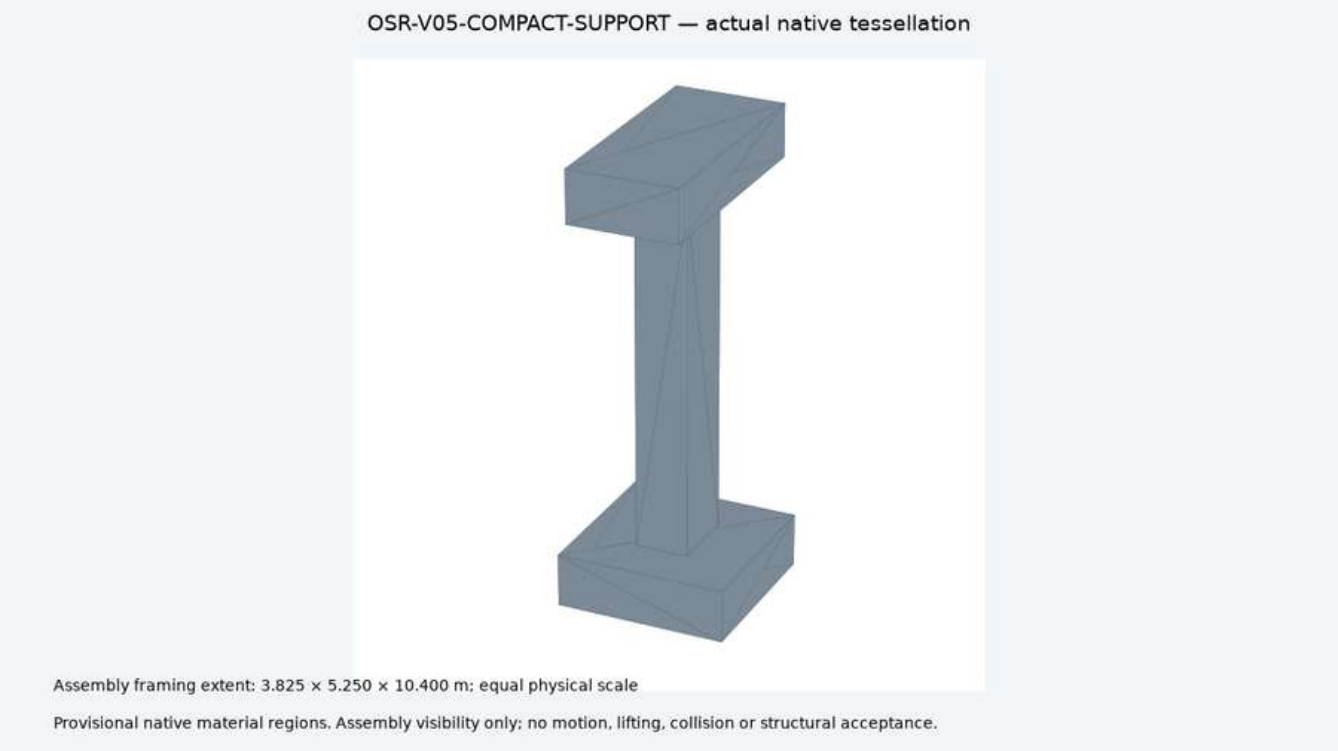
Beam-only civil variant. Supports are envelopes; evacuation acceptance remains open.

Beam-only civil variant. Supports are envelopes; evacuation acceptance remains open.



Partial evacuation entry frame and empty cassette. No deployed or stowed mechanism is qualified.

Partial evacuation entry frame and empty cassette. No deployed or stowed mechanism is qualified.



Compact pier and foundation envelope. Site capacity and structural acceptance remain open.

Compact pier and foundation envelope. Site capacity and structural acceptance remain open.

Primary maglev city directory

Separate planning packages. Complete programme costs, local content, land and physical acceptance remain open for every city.

City	Provisional trainsets	Cars	Open gates
Baghdad	550	3300	12
Benguela	115	345	12
Huambo	91	273	12
Lobito	64	192	12
Luanda	650	3900	12
Lubango	119	357	12
Malanje	28	84	12
Namibe	79	158	12
Soyo	63	126	12
Uige	17	51	12
Bukavu	118	354	12
Goma	85	255	12
Kananga	50	200	12
Kinshasa	1334	8004	12
Kisangani	64	256	12
Lubumbashi	192	768	12
Mbuji Mayi	132	528	12
Eldoret	133	399	12
Garissa	48	96	12
Kakamega	67	134	12
Kisii	50	100	12
Kisumu	79	237	12
Kitale	66	132	12
Machakos	64	128	12
Malindi	53	106	12
Meru Ke	64	128	12
Mombasa	359	1436	12
Nairobi	822	4932	12
Naivasha	86	172	12
Nakuru	84	252	12
Nyeri	57	114	12
Thika	98	294	12
Antananarivo	536	3216	12
Beira	69	207	12
Chimoio	54	162	12
Lichinga	28	56	12
Maputo	249	996	12
Nacala	77	154	12
Nampula	80	240	12
Pemba Mz	66	132	12
Quelimane	15	45	12
Tete	88	264	12
Xai Xai	40	80	12
Huye	73	146	12
Kigali	204	816	12
Rubavu	61	122	12
Mogadishu	232	928	12
Waw	45	90	12
Arusha	87	261	12
Dar Es Salaam	800	4800	12
Dodoma	80	240	12
Iringa	64	128	12
Kigoma	52	104	12
Mbeya	77	231	12
Morogoro	76	228	12
Moshi	57	114	12
Mwanza	251	1004	12

City	Provisional trainsets	Cars	Open gates
Shinyanga	90	180	12
Songea	19	38	12
Sumbawanga	35	70	12
Tabora	47	94	12
Tanga	83	249	12
Zanzibar City	83	249	12
Arua	67	134	12
Entebbe	78	156	12
Fort Portal	55	110	12
Gulu	129	387	12
Hoima	53	106	12
Jinja	92	184	12
Kampala	300	1200	12
Lira	88	176	12
Masaka	62	124	12
Mbale	64	128	12
Mbarara	89	267	12
Soroti	21	42	12
Lusaka	337	2022	12
Lyon	357	1428	12
La Paz	252	1008	12
Cuenca	101	303	12
San Salvador	294	1176	12
Agadir	120	360	12
Beni Mellal	46	92	12
Fez	191	764	12
Kenitra	105	315	12
Khouribga	39	78	12
Marrakech	322	1288	12
Meknes	82	246	12
Nador	67	134	12
Oujda	56	168	12
Safi	64	192	12
Tangier	247	988	12
Tetouan	98	294	12
El Obeid	81	243	12
Kassala	41	123	12
Khartoum	550	3300	12
Nyala	72	216	12
Omdurman	441	1764	12
Port Sudan	61	183	12
Tunis	275	1100	12
Bloemfontein	190	570	12
Durban	586	3516	12
East London Za	91	273	12
Nelspruit	57	114	12
Polokwane	83	249	12
Herat	64	192	12
Jalalabad Af	77	231	12
Kabul	466	2796	12
Kandahar	88	264	12
Mazar E Sharif	111	333	12
Barisal	108	324	12
Chittagong	519	3114	12
Comilla	150	450	12
Gazipur	311	1244	12
Khulna	255	1020	12
Mymensingh	85	255	12
Narayanganj	123	369	12
Rajshahi	112	336	12
Rangpur	70	210	12
Sylhet	81	243	12
Agra	159	636	12

City	Provisional trainsets	Cars	Open gates
Bhopal	235	940	12
Coimbatore	435	2610	12
Indore	412	2472	12
Jodhpur	176	704	12
Kanpur	378	2268	12
Lucknow	371	2226	12
Madurai	243	972	12
Meerut	154	616	12
Patna	398	1592	12
Raipur	154	616	12
Rajkot	133	532	12
Ranchi	283	1132	12
Vadodara	198	792	12
Varanasi	222	888	12
Vijayawada	206	824	12
Visakhapatnam	264	1056	12
Biratnagar	55	110	12
Kathmandu	373	1492	12
Pokhara	126	378	12
Bahawalpur	79	237	12
Faisalabad	235	1410	12
Gujranwala	207	828	12
Hyderabad Pk	296	1184	12
Karachi	689	4134	12
Larkana	50	150	12
Multan	164	656	12
Peshawar	204	816	12
Quetta	125	500	12
Rahim Yar Khan	99	297	12
Sheikhupura	58	174	12
Sialkot	83	249	12
Sukkur	73	219	12
Colombo	473	2838	12
Galle	99	297	12
Jaffna	95	285	12
Kandy	107	321	12
Phnom Penh	432	1728	12
Bandung	419	1676	12
Surabaya	469	2814	12
Vientiane	117	351	12
Mandalay	266	1064	12
Yangon	691	4146	12
Davao	309	1236	12
Ouagadougou	294	1176	12
Bafoussam	95	285	12
Bamenda	80	240	12
Bertoua	36	108	12
Douala	243	1458	12
Edea	11	22	12
Garoua	59	177	12
Kumba	62	186	12
Maroua	80	240	12
Ngaoundere	44	132	12
Yaounde	334	2004	12
Conakry	139	556	12
Bamako	413	1652	12
Niamey	233	932	12
Aba Ng	56	168	12
Benin City	230	920	12
Ibadan	259	1554	12
Ilorin	79	237	12
Jos	91	273	12
Kano	559	3354	12

City	Provisional trainsets	Cars	Open gates
Maiduguri	209	836	12
Onitsha	231	924	12
Port Harcourt	231	924	12
Uyo	63	189	12
Dakar	372	2232	12
Arish	24	48	12
Asyut	80	240	12
Beni Suef	55	165	12
Damanhur	82	246	12
Damietta	110	330	12
Fayoum	92	276	12
Hurghada	54	108	12
Ismailia	90	270	12
Kafr El Sheikh	60	120	12
Luxor	103	309	12
Mahalla	58	174	12
Mansoura Eg	75	225	12
Minya	89	267	12
Port Said	54	162	12
Qena	95	285	12
Sohag	84	252	12
Suez	89	267	12
Tanta	107	321	12
Zagazig	64	192	12
Amarah	76	228	12
Baqubah	110	330	12
Basra	399	2394	12
Diwaniyah	88	264	12
Duhok	86	258	12
Erbil	202	808	12
Fallujah	80	240	12
Hillah	95	285	12
Karbala	208	832	12
Kirkuk	197	788	12
Kut	70	210	12
Mosul	257	1028	12
Najaf	246	984	12
Nasiriyah	79	237	12
Ramadi	85	255	12
Samawah	84	252	12
Sulaymaniyah	121	484	12
Amman	440	2640	12
Aqaba	55	110	12
Irbid	76	228	12
Zarqa	103	309	12
Beirut	214	856	12
Sidon	54	108	12
Tripoli Lb	72	216	12
Gaza City	47	141	12
Hebron	108	324	12
Nablus	109	327	12
Abha	99	297	12
Al Kharj	98	294	12
Buraidah	100	300	12
Dammam	313	1252	12
Hail	116	348	12
Hofuf	111	333	12
Jeddah	491	2946	12
Jizan	90	270	12
Khamis Mushait	148	444	12
Mecca	252	1008	12
Medina	208	832	12
Najran	88	264	12

City	Provisional trainsets	Cars	Open gates
Tabuk	96	288	12
Taif	93	279	12
Aleppo	383	1532	12
Damascus	294	1176	12
Deir Ez Zor	90	270	12
Hama	111	333	12
Homs	68	204	12
Idlib	63	126	12
Latakia	90	270	12
Raqqa	69	207	12
Tartus	58	116	12
Aden	81	243	12
Dhamar	58	116	12
Hodeidah	48	144	12
Ibb	96	288	12
Lahij	49	98	12
Mukalla	122	366	12
Sanaa	405	2430	12
Sayun	42	84	12
Taiz	89	267	12